ADVANCED PYTHON PROGRAMMING

LAB ASSIGNMENT 9 – PROJECT REVIEW 2

ENCRYPTED DIARY: A BUSINESS THESIS PRESENTATION

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Encrypted Diary: Project Presentation Audio Link

https://drive.google.com/file/d/1h6p_4UXqJDPopjHquL3dJe5myN932__Y/view?usp=sharing

Encrypted Diary: Project Presentation Transcript

00:00

Hello, myself Monika and my project is called Encrypted Diary.

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So this project is mainly about protecting personal information. Like many people use digital diaries or WhatsApps to save their personal information. I mean, personal thoughts, daily experiences and other private information. But when you store this information online, there is always a chance that someone could access it without permission, right? Like most of the applications protect the user's account with a password. But here I wanted to add another layer of security to the actual diary account.

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That's the main idea behind this. So it's basically a website where users can create multiple diaries and each diary has its own password. The entries are encrypted before they are saved in the database so that the original content is not directly visible.

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So coming to the users and business workflow, the application is mainly designed for the people who want to keep their personal information private, like students, working professionals, and people who enjoy writing diaries.

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So first, users create an account and log into the website. After logging in, they can see their dashboard where they can manage their diaries and view the number of entries in each diary. Users can create different diaries for different purposes. For example, they can have one diary for daily experiences and another for personal notes.

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So each diary has a separate password which is different from the account password. So when a user wants to open a diary, they must enter its password. The application uses this password to generate an encryption key and unlock the diary. Once it is unlocked, the users can read the existing entries. Whenever a new entry is saved, its title and content are encrypted before storing, and then users can lock the diary again when it is finished.

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So coming to the technology stack and architecture, for this project, I've used Python, Flask, PostgreSQL, and Fernet encryption. So the frontend is built using HTML, CSS, and JavaScript, where HTML creates the page structure, CSS handles the design, and JavaScript adds the interactive features. Flask is used for the backend.

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It handles the main application functions such as registration, login, creating diaries and saving entries. PostgreSQL is used to store the user details, diary information and encrypted entries. And I have hosted the database using Neon, so I don't have to manage my own database servers.

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For security, account passwords are stored as hashes instead of plain text. For diary encryption, I've used PBKDF2 to generate an encryption key from the diary password and Fernet to encrypt and decrypt the diary entries.

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So the overall process is simple. Like the user interacts with the website, Flask processes the request and PostgreSQL stores the data. Before diary content is saved, it is encrypted. When the user unlocks the diary, the application decrypts the content so it can be read.

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I've used GitHub to manage the source code, Vercel to deploy the application, and Neon to host the database. Thank you.